SECOND ORDER

AI, Capital & Work · A Second Order research briefing

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The Ten-Year GPU

The idea circulating in AI credit is that disaggregated inference can move an old Hopper or Ampere GPU into a narrower role and stretch its life toward ten to fifteen years. Several vendors are building the pieces. Across five operators' filings, no disclosed GPU debt reaches year ten.

A working read of one claim, that new hardware extends the life of the last, followed from vendor materials and company filings and held open to correction.

IN THIS BRIEFING

Can new hardware extend an old GPU's life?

THE THROUGH-LINE

Since Issue 06 the series has followed the AI buildout from the power asset to the hardware that draws on it. This issue tests a claim now circulating in AI credit, raised by investor Gavin Baker on a recent podcast, that disaggregated inference, splitting a request into prefill and decode across different chips, lets an old GPU keep earning past the three-to-four-year life its credit was underwritten on.

WHY NOW

The architecture arrived in current-generation parts this year, in an AWS-Cerebras Bedrock system, NVIDIA's Rubin and Groq 3 LPX rack, and a cross-vendor research prototype. The financing has not followed. In five operators' filings, the disclosed GPU debt clocks end by year six.

WHAT YOU'LL LEARN

1. New hardware may extend old hardware, but one same-generation case is operating (§01)
2. The network and the site become part of the GPU's useful life (§02)
3. The disclosed GPU debt clocks end by year six (§03)
4. Management has moved depreciation lives toward five, and stopped (§04)
5. Complementary infrastructure can make compute cumulative (§05)
6. The next test is an old GPU disclosed inside a new system (§06)

IF YOU REMEMBER ONE LINE: the architecture that would extend an old GPU exists in new parts, but no disclosed financing underwrites years six through ten.

THE CLAIM

An old chip joined to a new system

> *The ten-year GPU is an old chip joined to a new system. The architecture is real but still unproven in the field, with no operator yet disclosing an old fleet inside such a system, and only IREN's debt cleanly matches its asset and its customer.*
>
> Second Order · Issue 07 claim

2.0x

combined H100 serving gain in one NVIDIA-Sarvam case; the disaggregation step alone about 1.5x.

2 to 6 yr

the disclosed GPU-linked debt maturities across five operators. None reaches year ten.

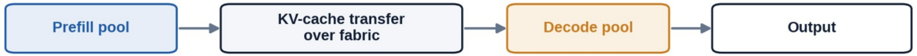
1 of 5

operators with a clean project-level match of asset life, customer term, and debt: IREN.

EXHIBIT 1

Phase separation is operating; old-vintage heterogeneous deployment remains unobserved

Disaggregated-inference evidence by maturity stage: who runs prefill, who runs decode, and what links them



Scheduler and router allocate requests across both pools

EVIDENCE	PREFILL	DECODE	TRANSFER & SYSTEM LAYER	WHAT IT PROVES
NVIDIA + Sarvam AI	H100	H100	SGLang router, 1P+1D KV transfer	Kernels, scheduling, and the split lifted one H100 case to 2.0x; the split alone about 1.5x
AWS + Cerebras	AWS Trainium	Cerebras CS-3	Amazon EFA; Bedrock, forthcoming	Commercial design assigns phases to different accelerator types
NVIDIA Rubin + Groq 3 LPX	Rubin GPU	Groq 3 LPU	Fabric logic, BlueField-4, liquid-cooled rack	Current product pairs GPUs, LPUs, network, and rack
HMA-Serve (preprint)	Tenstorrent GDDR	A100 80GB (old)	RDMA, cross-vendor KV-format conversion	Prototype puts an A100 80GB on decode; higher goodput per dollar
Life-extension claim	Hopper or Ampere	Cerebras or Groq LPU	Required, not quantified	Proposed 10 to 15 year vintage extension; no operator result

Note: NVIDIA-Sarvam uses H100 for both phases; the split alone is about 1.5x for the tested workload and latency target, and AWS's deployment guidance places the split's gains in long-context, high-concurrency serving (short prompts favor colocation). AWS-Cerebras is announced; Rubin-Groq pairs current components; HMA-Serve is a preprint that assigns an NVIDIA A100 80GB to decode, the reverse of the prefill role in the claim. Role assignment is not settled, and no row observes post-year-five Hopper or Ampere economics.

Source: NVIDIA; Cerebras and AWS; HMA-Serve; Gavin Baker (Invest Like the Best).

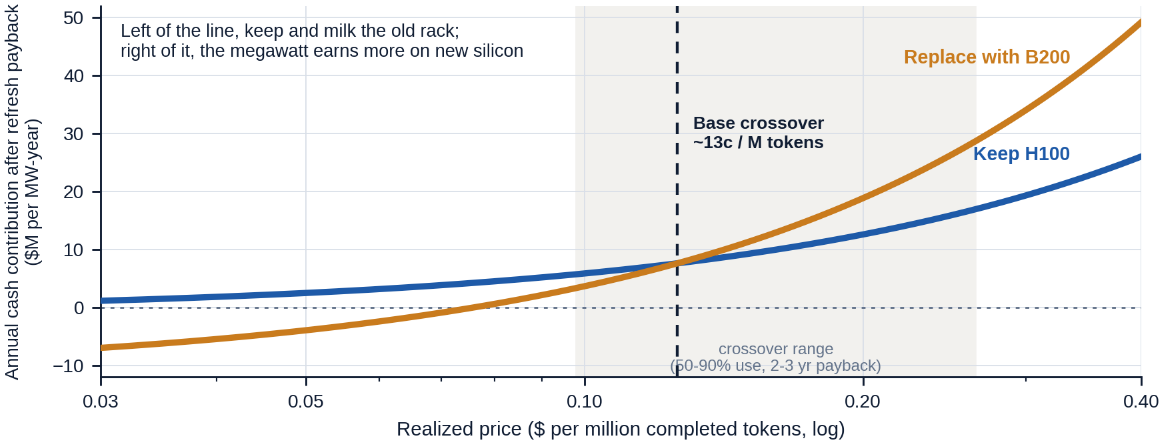
SO WHAT

- Phase separation is operating: combined tuning lifted one H100 case to 2.0x, and the disaggregation step alone about 1.5x.
- Which role the old card takes is unsettled: the one cross-vendor prototype assigns an NVIDIA A100 80GB to decode, and announced systems split the phases across different silicon.
- No public disclosure yet places a post-year-five Hopper or Ampere fleet inside such a system.

EXHIBIT 2

A profitable old rack can still lose a scarce megawatt

Annual cash contribution after refresh payback: keep an H100 rack vs. replace with B200, by realized price ($M per MW-year)



Note: Simple-payback sensitivity, no discounting. Base case: 1 MW, 70% use on both systems, power 8 plus 6 cents/kWh operating cost, 3-year payback at 32,000 dollars per kW; crossover 10 to 26 cents over 50 to 90% use and 2 to 3 year payback. Assumes the incremental B200 tokens sell at the marked price; the anchors (open-model output about 28 cents/M, market token index about 1.75 dollars/M) price unmatched workloads and are read for direction. Omits undisclosed integration capital, downtime, and salvage. Bounds, does not price.

Source: MLCommons Inference v5.0 Llama2-70B Server (DGX H100, B200), NVIDIA max input power; realized-price anchors DeepSeek and Silicon Data token index; Second Order scenario.

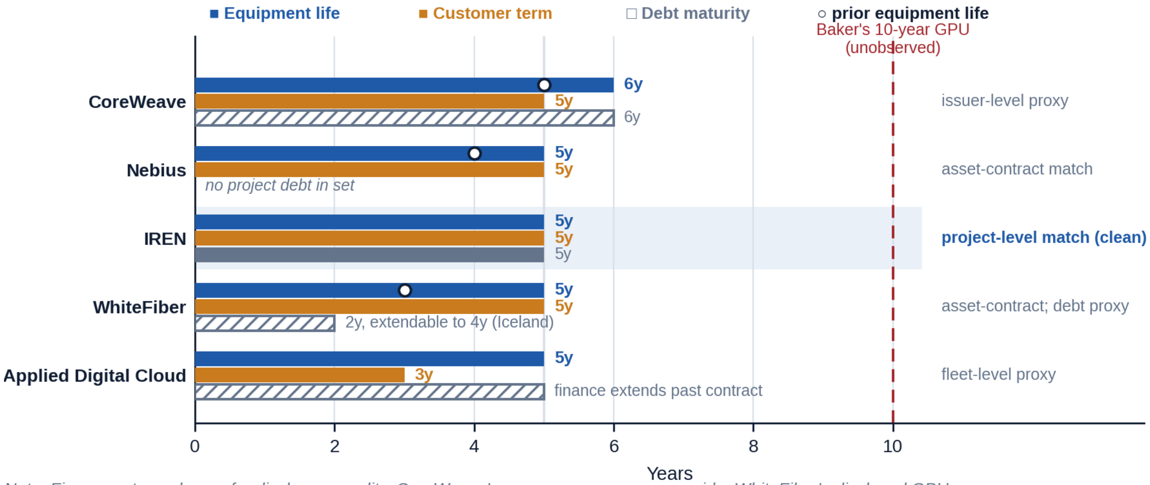
SO WHAT

- A retained Hopper rack stays cash-positive, but above about 13 cents per million tokens the megawatt earns more on Blackwell.
- That crossover sits at the floor of realized token prices, so a power-bound site leans to replacement; the old fleet earns where power is cheap.
- A labeled simple-payback sensitivity, not a live quote; it omits undisclosed integration capital, downtime, and salvage.

EXHIBIT 3

No disclosed GPU-linked maturity in this source set reaches year ten

Equipment life, customer commitment, and disclosed debt maturity, by operator (years)



Note: Five operators, chosen for disclosure quality. CoreWeave's measures are company-wide; WhiteFiber's disclosed GPU loan is a different fleet from its Paris contract; Applied Digital Cloud's leases are not mapped one-to-one to the named customer. Only IREN cleanly matches asset life, customer term, and project debt. Hatched debt bars are approximate or redacted. Hollow circles mark the prior disclosed equipment life; every revision moved toward five and none past six.

Source: CoreWeave, Nebius, IREN, WhiteFiber, and Applied Digital Cloud filings. Second Order calculations.

SO WHAT

- A GPU loan runs on three clocks: equipment life, customer term, and debt tenor; across five operators, lives cluster at five to six years, terms at three to five, and debt at two to six.
- Hollow circles mark prior lives: every revision moved toward five, none past six, and lengthening the life lifts earnings, CoreWeave by 20 million dollars, Nebius by 43.1 million.
- Only IREN cleanly matches all three clocks, and its debt ends no later than the Microsoft service fees.

Takeaways

1. Useful life is now a systems question

An old GPU's later value depends on the complementary accelerator, the network, the serving layer, and the site as much as on the chip.

2. The financing has not followed the architecture

The pieces exist in current-generation parts, and no disclosed senior debt underwrites years six through ten.

3. Five years is management's honest horizon

Accounting lives and customer terms cluster near five, and the ten-year system has yet to appear in any filing.

4. Advance rates first, spreads later

Proven residual raises the advance rate and splits GPU credit into a contract tranche and a residual tranche; spread cuts come later.

WATCHLIST

What would change our mind

The first vintage disclosure: an operator naming an older fleet moved into a heterogeneous system beside newer hardware, on prefill or decode, with the throughput, power, and realized revenue before and after.

A post-year-five refinancing: the same aging fleet refinanced against a renewed customer contract and a disclosed heterogeneous architecture.

A renewed customer term beyond year five: which would extend contracted revenue visibility into the later period.

The falsifier: operators deploy heterogeneous systems and still retire Hopper and Ampere on the old schedule, or the enabling capital costs more than the cash flow it preserves. Either would send us back to reevaluate the thesis.

THE SECOND-ORDER EFFECT · LONGER LIFE BUILDS A RESIDUAL MARKET BEFORE IT CUTS SPREADS. Proven residual raises advance rates before it cuts coupons and splits GPU credit into a contract and a residual tranche. The deeper end point: GPU financing as an asset class, with residual indices, securitizations, and retrofit reserves, like aircraft finance.